Lesson 08 Demo 02

Running Multiple Cloud Provider Configurations

Objective: To demonstrate the use of multiple AWS provider configurations in Terraform for managing resources across different regions, showcasing the ability to define, configure, and deploy resources in various regions within a single Terraform configuration

Tools required: VS Code and Linux terminal

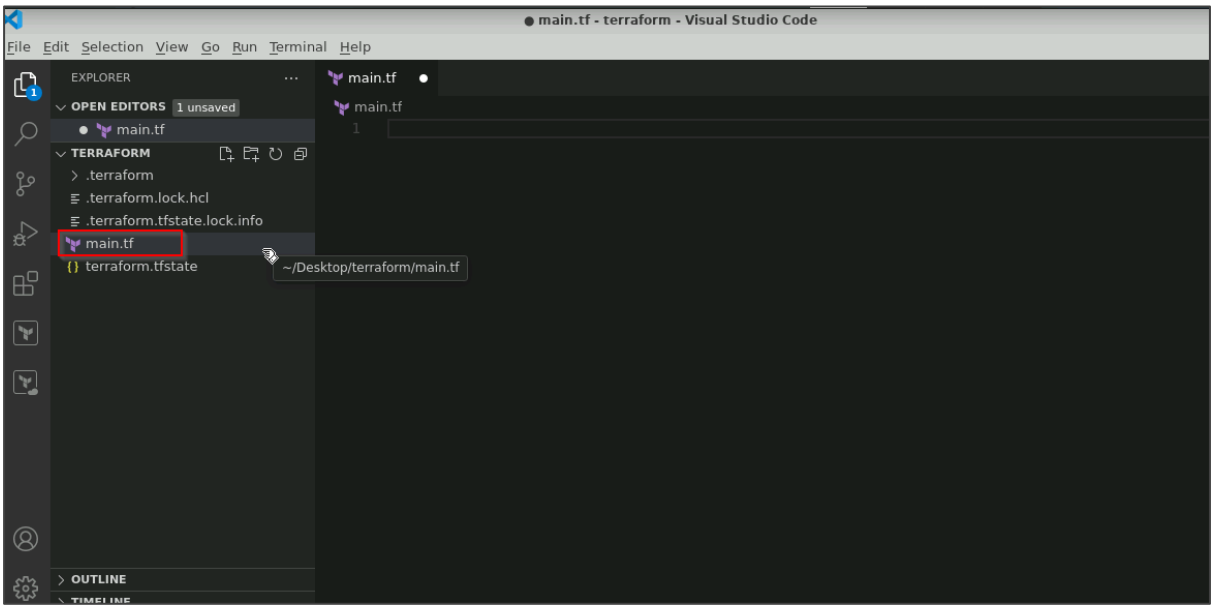
Prerequisites: None

Steps to be followed:

1. Define providers with aliases in the **main.tf** file
2. Specify provider configurations for each resource in the **main.tf** file
3. Initialize and apply the configuration

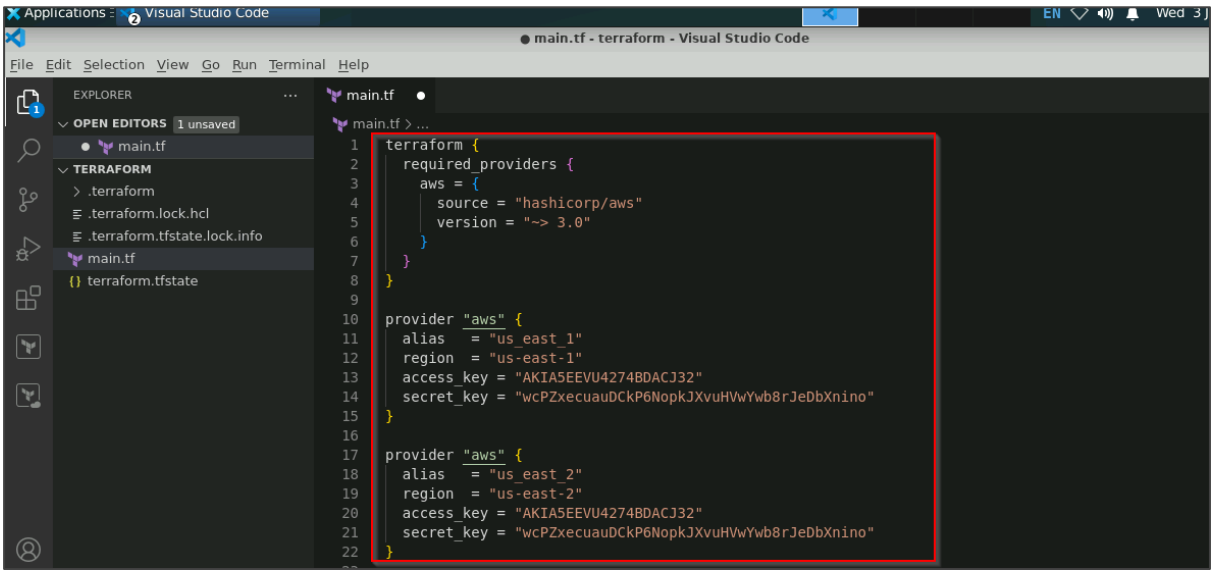
Step 1: Define providers with aliases in the main.tf file

1.1 Open the **main.tf** file in the **Terraform** folder



1.2 Enter the code given below in the **main.tf** file to define multiple AWS providers with different aliases and save the file:

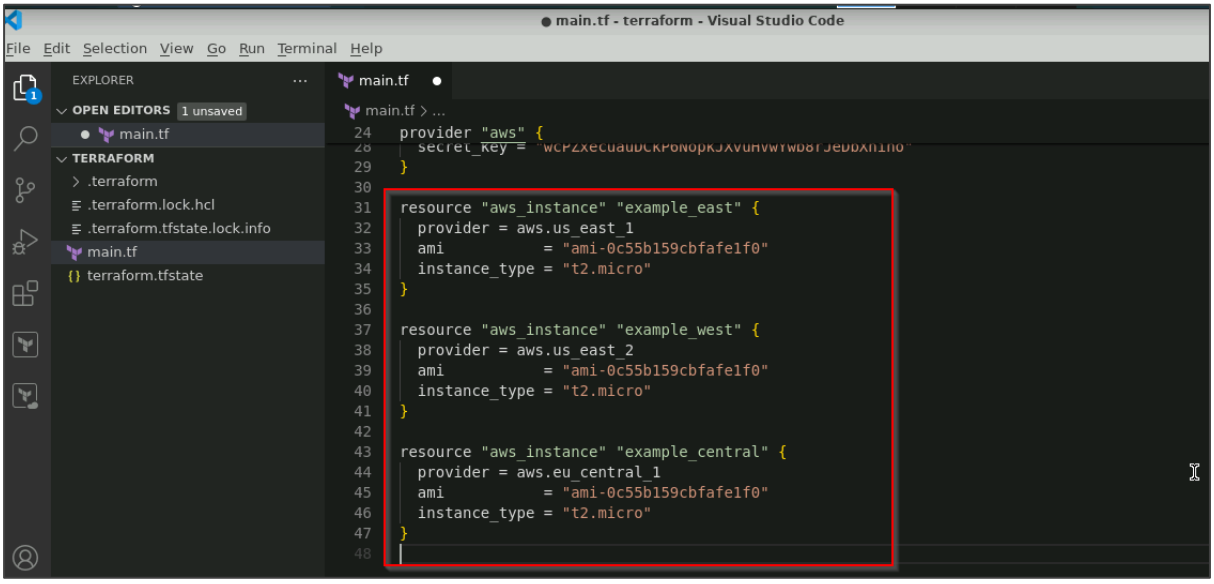
```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 3.0"
    }
  }
}

provider "aws" {
  alias   = "us_east_1"
  region  = "us-east-1"
  access_key = "AKIA5EEVU4274BDACJ32"
  secret_key = "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"
}

provider "aws" {
  alias   = "us_east_2"
  region  = "us-east-2"
  access_key = "AKIA5EEVU4274BDACJ32"
  secret_key = "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"
}

provider "aws" {
  alias   = "eu_central_1"
  region  = "eu-central-1"
  access_key = "AKIA5EEVU4274BDACJ32"
  secret_key = "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"
}
```



Step 2: Specify provider configurations for each resource in the main.tf file

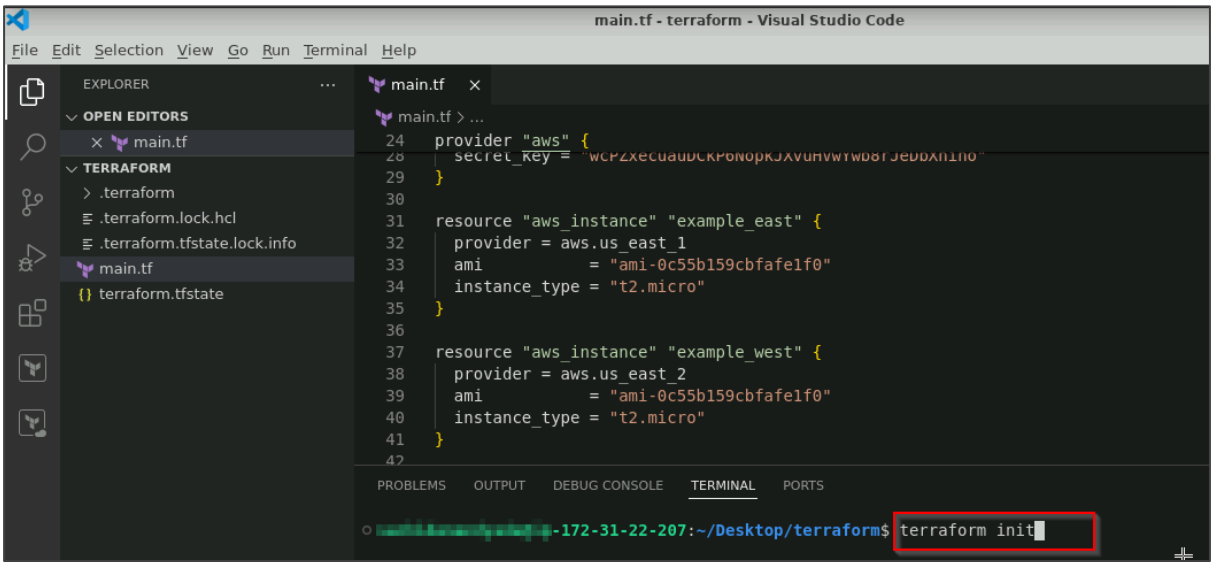
2.1 Update the **main.tf** file with the provider configurations for each resource as given below and save the file:

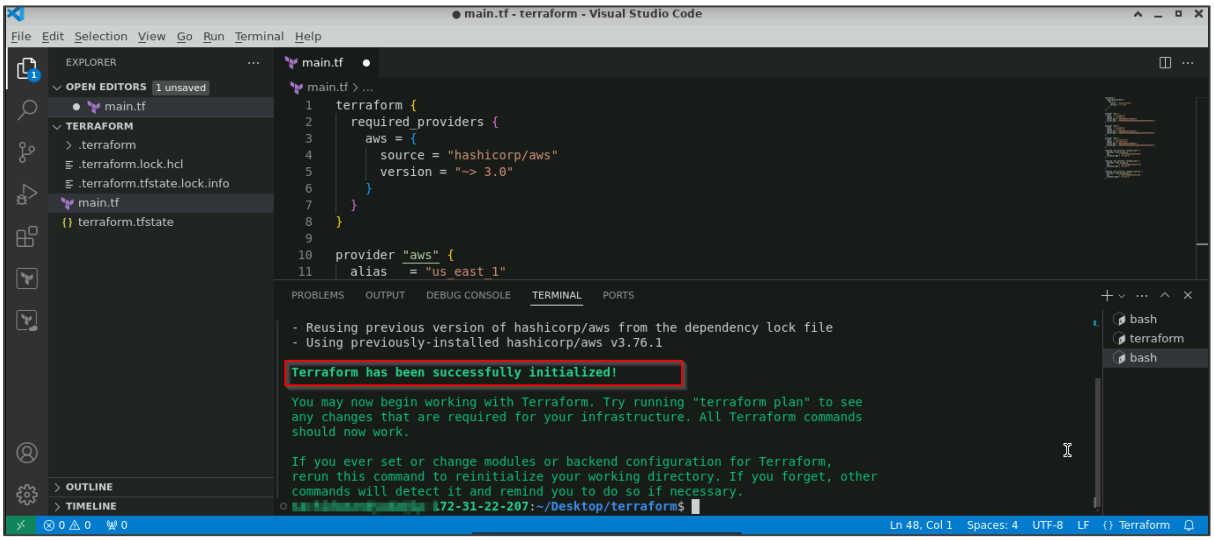
```
resource "aws_instance" "example_east" {
  provider = aws.us_east_1
  ami          = "ami-0c55b159cbfafe1f0"
  instance_type = "t2.micro"
}

resource "aws_instance" "example_west" {
  provider = aws.us_east_2
  ami          = "ami-0c55b159cbfafe1f0"
  instance_type = "t2.micro"
}

resource "aws_instance" "example_central" {
  provider = aws.eu_central_1
  ami          = "ami-0c55b159cbfafe1f0"
  instance_type = "t2.micro"
}
```



Step 3: Initialize and apply the configuration

3.1 Open the terminal and run the command given below to initialize the configurations:
terraform init

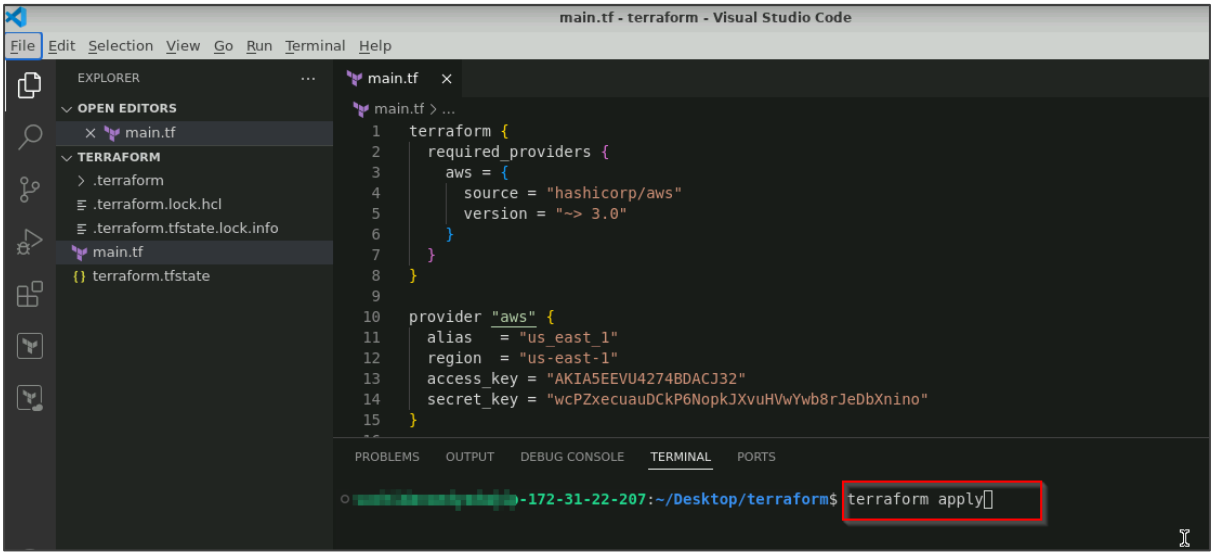




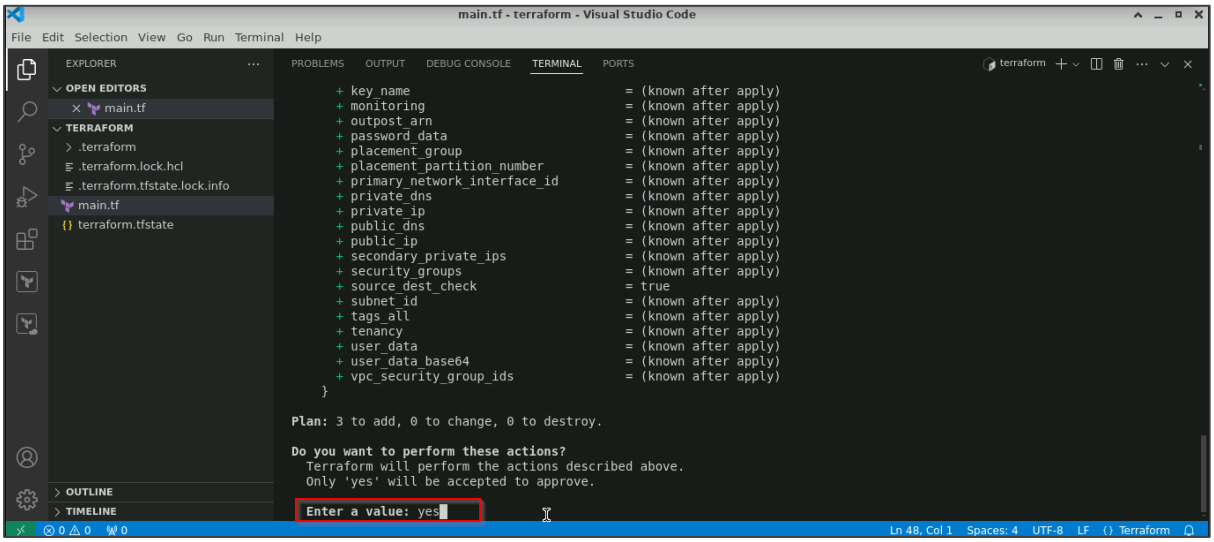
The Terraform configuration is successfully initialized.

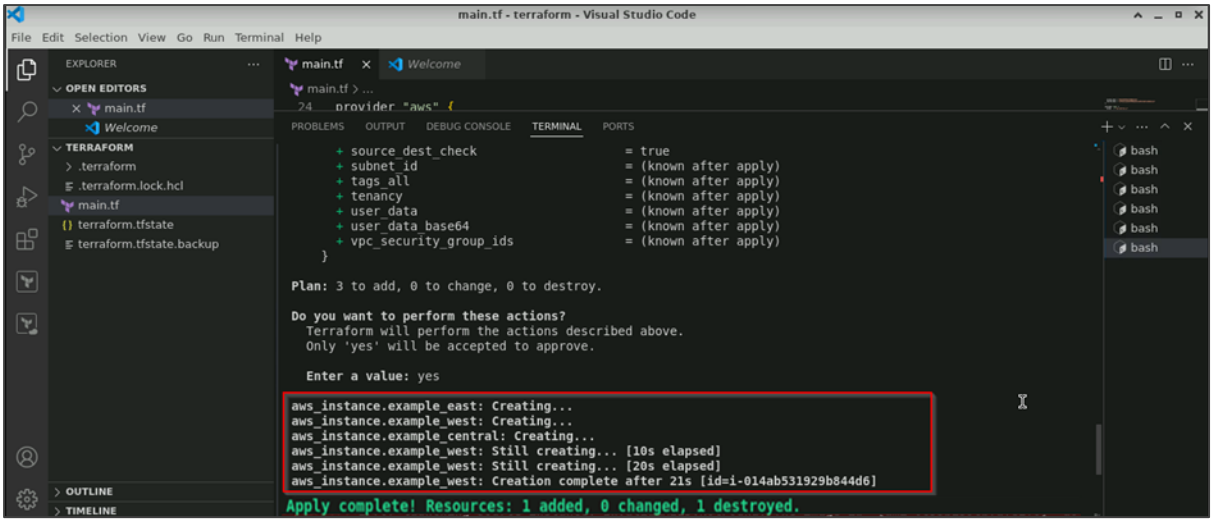
3.2 Open the terminal and run the command given below to apply the configurations:

terraform apply



3.3 Enter **yes** to apply the configuration





By following these steps, you have successfully demonstrated the use of multiple AWS provider configurations in Terraform for managing resources across different regions, showcasing the ability to define, configure, and deploy resources in various regions within a single Terraform configuration.